

If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

- a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.
- b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.
- c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.
- d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).
- e. Bonus problems are not required for credit, but they are worth extra credit.

Definitions & Facts. A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is **convex** if and only if for all x, y the inequality holds:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad (\text{first-order condition}).$$

If f is twice continuously differentiable, a sufficient (and, on open convex domains, also necessary) condition is

$$\nabla^2 f(x) \succeq 0 \quad \text{for all } x \quad (\text{Hessian positive semidefinite}).$$

If $\nabla^2 f(x) \succ 0$ everywhere, then f is *strictly* (indeed, strongly) convex. Recall that a matrix A is positive semidefinite if

$$v^\top A v \geq 0 \quad \forall v \in \mathbb{R}^d,$$

and is positive definite if the inequality is strict. Moreover, as we showed in class, a matrix is positive semidefinite if and only if all its eigenvalues are non-negative.

For a square matrix $A \in \mathbb{R}^{n \times n}$, the **trace** of A , denoted $\text{tr}(A)$, is the sum of its diagonal entries:

$$\text{tr}(A) = \sum_{i=1}^n A_{ii}.$$

Useful properties. For any conformable matrices A, B, C :

$$\begin{aligned} \text{tr}(A + B) &= \text{tr}(A) + \text{tr}(B), \\ \text{tr}(kA) &= k \text{tr}(A) \quad \text{for scalar } k, \\ \text{tr}(A^\top) &= \text{tr}(A), \\ \text{tr}(AB) &= \text{tr}(BA) \quad (\text{cyclic property}), \\ \text{tr}(ABC) &= \text{tr}(BCA) = \text{tr}(CAB). \end{aligned}$$

These identities are widely used when differentiating matrix functions, especially expressions like $\text{tr}(A^\top X)$ or $\text{tr}(X^\top A X)$.

Problems

Problem 1. (Quadratic Function with Linear Term.)

Consider

$$f(x) = \frac{1}{2}x^\top Qx + q^\top x + r, \quad Q = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \quad q = \begin{bmatrix} -2 \\ -8 \end{bmatrix}.$$

- (a) Derive expressions for $\nabla f(x)$ and H .
- (b) Find the minimizer x^* .
- (c) Compute $f(x^*)$.
- (d) Sketch or describe the level sets of f . Are they circles or ellipses? Why?

Solution.
(a) Gradient and Hessian.

We have

$$f(x) = \frac{1}{2}x^\top Qx + q^\top x + r.$$

For a quadratic function of this form:

$$\nabla f(x) = Qx + q, \quad \nabla^2 f(x) = Q.$$

Substituting the given data,

$$Q = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \quad q = \begin{bmatrix} -2 \\ -8 \end{bmatrix},$$

we get

$$\nabla f(x) = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} -2 \\ -8 \end{bmatrix} = \begin{bmatrix} 2x_1 - 2 \\ 4x_2 - 8 \end{bmatrix}, \quad H = \nabla^2 f(x) = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}.$$

(b) Minimizer.

Set the gradient to zero:

$$\nabla f(x^*) = 0 \quad \Rightarrow \quad Qx^* + q = 0 \quad \Rightarrow \quad x^* = -Q^{-1}q.$$

Compute:

$$Q^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{bmatrix}, \quad x^* = - \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{bmatrix} \begin{bmatrix} -2 \\ -8 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

(c) Value at the minimum.

$$f(x^*) = \frac{1}{2}(x^*)^\top Qx^* + q^\top x^* + r.$$

Substitute:

$$(x^*)^\top Qx^* = [1 \ 2] \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = [1 \ 2] \begin{bmatrix} 2 \\ 8 \end{bmatrix} = 2 + 16 = 18.$$

and

$$q^\top x^* = [-2 \ -8] \begin{bmatrix} 1 \\ 2 \end{bmatrix} = -2 - 16 = -18.$$

Therefore,

$$f(x^*) = \frac{1}{2}(18) - 18 + r = 9 - 18 + r = -9 + r.$$

(d) Level sets.

The quadratic term defines level sets of

$$\frac{1}{2}x^\top Qx + q^\top x = \text{constant}.$$

Since Q is symmetric positive definite and diagonal with unequal entries (2 and 4), the level sets are *ellipses*, not circles. They are centered at $x^* = (1, 2)$, with principal axes aligned with the coordinate axes and axis lengths inversely proportional to the square roots of the eigenvalues of Q .

Level sets: ellipses centered at $(1, 2)$, elongated more in the x_2 direction.

Problem 2. (Deriving $\nabla_X \text{tr}(A^\top X)$ from first principles.)

Let $A, X \in \mathbb{R}^{m \times n}$ and define

$$f(X) = \text{tr}(A^\top X).$$

- (a.) Write $f(X)$ out as a double sum over entries of A and X .
- (b.) Compute the partial derivative $\frac{\partial f}{\partial X_{ij}}$ directly from that expression.
- (c.) Assemble your results into a matrix to find $\nabla_X f(X)$.
- (d.) Verify your result by checking a simple case where $m = n = 2$ and A and X have arbitrary entries.
- (e.) Use the result to find $\nabla_X \text{tr}(X^\top A)$.

Solution.

(a) Recall that $\text{tr}(A^\top X)$ is the sum of diagonal entries of $A^\top X$. In particular, we have that

$$(A^\top X)_{ii} = \sum_{k=1}^m (A^\top)_{ik} X_{ki}.$$

We also know that $(A^\top)_{ik} = A_{ki}$ so then

$$\text{tr}(A^\top X) = \sum_{i=1}^n (A^\top X)_{ii} = \sum_{i=1}^n \sum_{j=1}^m A_{ji} X_{ji}$$

(b) Take the partial derivative with respect to X_{kl} :

$$\frac{\partial f}{\partial X_{kl}} = A_{lk},$$

because only the term with $i = k, j = l$ survives the sum.

(c) Form the gradient matrix whose (k, l) entry is $\frac{\partial f}{\partial X_{kl}}$:

$$(\nabla_X f(X))_{kl} = A_{lk}.$$

This means

$$\nabla_X f(X) = A.$$

Notice the transpose pattern: the derivative flips the order inside $\text{tr}(A^\top X)$.

(d) For a 2×2 example, if $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, $X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$, then

$$f(X) = a_{11}x_{11} + a_{21}x_{12} + a_{12}x_{21} + a_{22}x_{22},$$

so clearly $\frac{\partial f}{\partial x_{11}} = a_{11}$, $\frac{\partial f}{\partial x_{12}} = a_{21}$, etc., giving $\nabla_X f = A$.

(e) Since $\text{tr}(X^\top A) = \text{tr}(A^\top X)$, the same result holds:

$$\boxed{\nabla_X \text{tr}(X^\top A) = A.}$$

Problem 3. (Matrix derivatives.) Throughout, use the rule that $\nabla_X \text{tr}(A^\top X) = A$ and linearity of differentiation. All variables are real-valued matrices.

- (a) Let $f(X) = \text{tr}(A^\top X)$. Derive an expression for $\nabla_X f(X)$.
- (b) Let $f(X) = \text{tr}(X^\top A)$. Derive an expression for $\nabla_X f(X)$.
- (c) Let $f(X) = \text{tr}(AXB)$. Derive an expression for $\nabla_X f(X)$.
- (d) Let $f(X) = \text{tr}(AX^\top B)$. Derive an expression for $\nabla_X f(X)$.

Solution.

(a) $\text{tr}(A^\top X) = \sum_{ij} A_{ij} X_{ij}$, so $\frac{\partial f}{\partial x_{ij}} = A_{ij}$. Thus $\nabla_X f = A$.

(b) $\text{tr}(X^\top A) = \text{tr}(A^\top X)$, so $\nabla_X f = A$ again.

(c) $\text{tr}(AXB) = \text{tr}(BAX)$ by cyclic permutation, so $\nabla_X f = A^\top B^\top = A^\top B^\top$. More neatly,

$$\boxed{\nabla_X f = A^\top B^\top.}$$

(d) $\text{tr}(AX^\top B) = \text{tr}(BAX^\top) = \text{tr}(X^\top(BA))$, so

$$\boxed{\nabla_X f = (BA)^\top = A^\top B^\top.}$$

Problem 4. (Least Angle Property of Least Squares.) Suppose the matrix $X \in \mathbb{R}^{m \times n}$ has linearly independent columns, and let $y \in \mathbb{R}^m$. Let $\hat{\theta}$ denote the least-squares approximate solution of $X\theta = y$, i.e.

$$(X^\top X)\hat{\theta} = X^\top y.$$

We will establish several geometric properties of the least-squares solution.

a. **Inner product property.** Show that for any $\theta \in \mathbb{R}^n$,

$$(X\theta)^\top y = (X\theta)^\top (X\hat{\theta}),$$

i.e., the inner product of $X\theta$ and y is the same as that of $X\theta$ and $X\hat{\theta}$. (Hint: use $(X\theta)^\top y = \theta^\top (X^\top y)$ and $(X^\top X)\hat{\theta} = X^\top y$.)

b. **Angle between $X\hat{\theta}$ and y .** Show that when both $X\hat{\theta}$ and y are nonzero,

$$\frac{(X\hat{\theta})^\top y}{\|X\hat{\theta}\| \|y\|} = \frac{\|X\hat{\theta}\|}{\|y\|}.$$

The left-hand side is the cosine of the angle between $X\hat{\theta}$ and y . Hence $X\hat{\theta}$ and y form the smallest possible angle.

c. **Least angle property of least squares.** The choice $\theta = \hat{\theta}$ minimizes the distance between $X\theta$ and y . Show that $\theta = \hat{\theta}$ also minimizes the angle between $X\theta$ and y . (You may assume $X\theta$ and y are nonzero.) *Remark.* For any scalar $\alpha > 0$, $\theta = \alpha\hat{\theta}$ also minimizes the angle between $X\theta$ and y .

Solution.

Denote the range space of X as

$$\mathcal{R}(X) := \{\zeta \mid X\xi = \zeta \text{ for some } \xi \in \mathbb{R}^n\} = \text{span}\{X\xi \mid \xi \in \mathbb{R}^n\} = \text{col}(X).$$

All of these after $:=$ are equivalent definitions of the range space of X . Let $X \in \mathbb{R}^{m \times n}$ have linearly independent columns and let $\hat{\theta} = X^\dagger y$ —where $X^\dagger = X^\top (XX^\top)^{-1}$ is the pseudo-inverse of X —be the least-squares solution of $X\theta \approx y$. Equivalently, $\hat{\theta}$ satisfies the normal equations

$$(X^\top X)\hat{\theta} = X^\top y.$$

Let $p := X\hat{\theta}$ (the projection of y onto $\mathcal{R}(X)$) and $r := y - p$ (the residual). Then $r \perp \mathcal{R}(X)$ (i.e. r is orthogonal to $\mathcal{R}(X)$), which means (by definition of orthogonality) that $X^\top r = 0$.

a. **Inner product property.** For any $\theta \in \mathbb{R}^n$,

$$(X\theta)^\top y = \theta^\top (X^\top y) = \theta^\top (X^\top X)\hat{\theta} = (X\theta)^\top (X\hat{\theta}).$$

Thus the inner product of $X\theta$ with y equals the inner product of $X\theta$ with $X\hat{\theta}$.

b. **Angle between $X\hat{\theta}$ and y .** Assume $X\hat{\theta} \neq 0$ and $y \neq 0$. Using part (a) with $\theta = \hat{\theta}$,

$$(X\hat{\theta})^\top y = (X\hat{\theta})^\top (X\hat{\theta}) = \|X\hat{\theta}\|^2.$$

Therefore

$$\frac{(X\hat{\theta})^\top y}{\|X\hat{\theta}\| \|y\|} = \frac{\|X\hat{\theta}\|^2}{\|X\hat{\theta}\| \|y\|} = \frac{\|X\hat{\theta}\|}{\|y\|}.$$

The left-hand side is $\cos \angle(X\hat{\theta}, y)$.

c. **Least angle property.** What we need to show is that $\theta = \hat{\theta}$ minimizes the angle between $X\theta$ and y . The angle between $X\theta$ and y is given by

$$\cos \angle(X\theta, y) = \frac{X\theta^\top y}{\|X\theta\| \|y\|}.$$

We want to show that this is minimized when $X\theta = X\hat{\theta}$.

Because $\hat{\theta}$ solves least squares, $p = X\hat{\theta}$ is the orthogonal projection of y onto the column space $\mathcal{R}(X)$, and $r = y - p$ is orthogonal to $\mathcal{R}(X)$. Indeed,

$$X^\top(y - p) = X^\top(y - X\hat{\theta}) = X^\top y - X^\top X\hat{\theta} = 0$$

since $\hat{\theta}$ solves the normal equations.

For any θ let $w := X\theta \in \mathcal{R}(X)$. Since $r \perp \mathcal{R}(X)$,

$$w^\top y = w^\top(p + r) = w^\top p.$$

By Cauchy–Schwarz (i.e. $\langle u, v \rangle \leq \|u\| \|v\|$), we have that

$$\frac{w^\top y}{\|w\| \|y\|} = \frac{w^\top p}{\|w\| \|y\|} \leq \frac{\|w\| \|p\|}{\|w\| \|y\|} = \frac{\|p\|}{\|y\|} = \frac{\|X\hat{\theta}\|}{\|y\|}.$$

Therefore

$$\cos \angle(X\theta, y) \leq \frac{\|X\hat{\theta}\|}{\|y\|}.$$

which from part b, is the same as the cosine of the angle between $X\hat{\theta}$ and y .

Equality holds if and only if w is colinear with p , i.e., $X\theta = \alpha X\hat{\theta}$ for some $\alpha > 0$. Thus the cosine of the angle between $X\theta$ and y is maximized (the angle minimized) precisely when $X\theta$ is parallel to $X\hat{\theta}$, with the maximum cosine $\|X\hat{\theta}\|/\|y\|$. In particular, taking $\theta = \hat{\theta}$ gives the minimum angle.

Remarks.

1. For any scalar $\alpha > 0$, $\theta = \alpha\hat{\theta}$ yields

$$\cos \angle(X(\alpha\hat{\theta}), y) = \frac{(\alpha X\hat{\theta})^\top y}{\|\alpha X\hat{\theta}\| \|y\|} = \frac{\alpha \|X\hat{\theta}\|^2}{|\alpha| \|X\hat{\theta}\| \|y\|} = \frac{\|X\hat{\theta}\|}{\|y\|},$$

so any positive rescaling of $\hat{\theta}$ achieves the same minimal angle.

2. The function $\arccos$ is strictly decreasing on $[-1, 1]$, so that means

$$\xi_1 > \xi_2 \iff \arccos(\xi_1) < \arccos(\xi_2).$$

In other words, the smaller the cosine of the angle, the larger the angle. Thus this is the maximal possible angle.

Problem 5. (Linear Functions (8.6 VMLS).) For each description of y , express it as $y = Ax$ for some A .

- (a) y_i is the difference between x_i and the average of $x_1, \dots, x_{i-1}$ (with $y_1 = x_1$).
- (b) y_i is the difference between x_i and the average of all the other entries, i.e., the average of $\{x_j : j \neq i\}$.

Solution. We express each description as $y = Ax$ for some matrix A .

(a) Difference from the average of previous entries. For $i = 1$, define $y_1 = x_1$. For $i \geq 2$,

$$y_i = x_i - \frac{1}{i-1} \sum_{j=1}^{i-1} x_j.$$

Hence, $A \in \mathbb{R}^{n \times n}$ is lower-triangular with

$$A_{ii} = 1, \quad A_{ij} = \begin{cases} -\frac{1}{i-1}, & j < i, \\ 0, & j > i, \end{cases}$$

and the first row $A_{1,1} = 1$, $A_{1,j} = 0$ for $j > 1$. **(b) Difference from the average of all other entries.** We want

$$y_i = x_i - \frac{1}{n-1} \sum_{j \neq i} x_j = x_i - \frac{1}{n-1} (\mathbf{1}^\top x - x_i) = \frac{n}{n-1} x_i - \frac{1}{n-1} \sum_{j=1}^n x_j.$$

Thus each row i has the same coefficients, giving

$$A = \frac{n}{n-1} I - \frac{1}{n-1} \mathbf{1}\mathbf{1}^\top.$$

Then $y = Ax$ satisfies the required relationship for all i .

Problem 6. (Weighted least squares.) Let $W \in \mathbb{R}^{n \times n}$ be a symmetric positive definite matrix and define

$$f(\theta) = (X\theta - y)^\top W(X\theta - y).$$

- Show that $\nabla_\theta f(\theta) = 2X^\top W(X\theta - y)$.
- Derive the normal equations—i.e., c, d such that $d = X^\top c$. Under what conditions can you find an expression for the minimizer θ^* , and what is the expression?
- Explain in words how the weighting matrix W affects which data points have more influence.

Solution.

(a) View $f(\theta) = (X\theta - y)^\top W(X\theta - y)$ with $W^\top = W$:

$$\nabla_\theta f(\theta) = 2X^\top W(X\theta - y).$$

(b) First-order condition: $X^\top W(X\theta^* - y) = 0$, i.e.

$$(X^\top W X) \theta^* = X^\top W y.$$

If $X^\top W X$ is invertible, then

$$\theta^* = (X^\top W X)^{-1} X^\top W y.$$

(c) Intuition: $W \succ 0$ reweights residuals. If $W = \text{diag}(w_i)$, then larger w_i give observation i more influence in fitting (its residual is penalized more). For general $W = S^\top S$ (Cholesky), the problem is equivalent to ordinary least squares on transformed data (SX, Sy) , i.e., a “prewhitening” of the residuals.

Problem 7. (Composition rule practice.) Let $f(x) = g(Ax + b)$ where $g : \mathbb{R}^m \rightarrow \mathbb{R}$ is differentiable and $A \in \mathbb{R}^{m \times n}$.

- (a) Show that $\nabla_x f(x) = A^\top \nabla g(Ax + b)$.
- (b) If $g(z) = \frac{1}{2} \|z\|_2^2$, compute $\nabla_x f(x)$ and $\nabla_x^2 f(x)$ explicitly.
- (c) Explain why this captures the gradient and Hessian of $\frac{1}{2} \|Ax + b\|_2^2$ without expanding coordinates.

Solution. (a) By the chain rule, with $z = Ax + b$,

$$\nabla_x f(x) = A^\top \nabla g(z) \Big|_{z=Ax+b}.$$

(b) If $g(z) = \frac{1}{2} \|z\|_2^2$, then $\nabla g(z) = z$ and $\nabla^2 g(z) = I$. Hence

$$\nabla_x f(x) = A^\top (Ax + b), \quad \nabla_x^2 f(x) = A^\top A.$$

(c) The expression $\frac{1}{2} \|Ax + b\|_2^2$ is the composition of a quadratic with an affine map; the chain rule yields the gradient/Hessian in compact matrix form without expanding coordinates.